

Jane Li

jane.sy.li@mail.mcgill.ca
jane.li@mila.quebec
(Last updated Oct 1 2026)

Research Interests

- ◇ Neuro-symbolic linguistic representations and biases in computational learners of language.
- ◇ Morpho-phonological representations employed during language production planning.

Education

Johns Hopkins University

PhD in Cognitive Science

advised by Kyle Rawlins (primary) and Paul Smolensky (secondary)

Aug 2021 – Present

Baltimore, MD, USA

Simon Fraser University

BA Honours in Linguistics, Minor in Philosophy

Sept 2017 – May 2021

Burnaby, BC, Canada

Publications / Conference proceedings / Preprints

Propositionality compositionality in neural language models

Jane Li, Abhinav Patil, & Kyle Rawlins

SCiL 2026

Linear representations of grammaticality in neural language models

Jane Li & Najoung Kim

Preprint

(R)NNs too expressive? No relationship between neural architectures' theoretical expressivity and empirical performance
Jane Li* & Alan Zhou*

Linguistics Vanguard
2026

CNNs that robustly compute vowel harmony do not explicitly represent phonological tiers

Jane Li & Alan Zhou

SCiL 2025

Frequency norms in Tashlhiyt

John Alderete, Jane Li, & Rachid Ridouane.

Handbook of Berber Linguistics
2024

Word and sound frequency norms in Cantonese: Comparisons across three corpora

Jane Li, John Alderete, & Heikal Badrulhisham.

Taiwan Journal of Linguistics
2023

**Leveraging paradigmatic information in inflection
acceptability prediction**

Jane Li & Colin Wilson. (Shared task paper)

SIGMORPHON
2021

**Were we there already? Applying minimal generalization to
the SIGMORPHON shared task on cognitively plausible
morphological inflection**

Colin Wilson & Jane Li. (Shared task paper)

SIGMORPHON
2021

Selected presentations

Talks

Representations of grammaticality in neural language models

Invited talk at BU tinlab (Oct 2025), GenLM reading group (Nov 2025).

**A widespread failure to represent feature-based phonological dependencies in neural
learners of phonology**

Invited talk at USC Phonology lab (2025).

**Do DNNs implicitly recover feature-like representations to represent
morpho-phonological processes?**

Invited talk at LSA Organized Session on Computational models of learnability and acquisition of morphology and phonology (2025).

A semantic and morphological analysis of Tagalog plural pronoun constructions

with Gérard Avelino.

At Austronesian Formal Linguistics Association, invited talk at Rutgers Linguistics (2024).

RNNs can correctly inflect while not learning human-like morpho-phonology

with Kyle Rawlins and Paul Smolensky. At Penn Linguistics Conference (2024).

**Dog[z], cat[s], and leash[iz]: processing differences among regular allomorphs during
speech production**

with Colin Wilson. At Architectures and Mechanisms for Language Processing (2023).

Recent posters (* denotes mentee)

CNNs that robustly compute vowel harmony do not explicitly represent tiers

with Alan Zhou. At Society for Computation in Linguistics (2025).

**Wug[z], wup[s], and wuz[iz]: disambiguating stem and allomorph effects in speeded
nonword inflection**

with Sofia Rest* and Colin Wilson. At Human Sentence Processing Conference (2025).

Modeling morphological production with an algorithmically specified InFLACT-R

with Grusha Prasad. At Human Sentence Processing Conference (2025).

Are learned “features” used as features? A case study on Turkish vowel harmony
with Kyle Rawlins and Paul Smolensky. At Society for Computation in Linguistics (2024).

Dog[z], cat[s], and leash[iz]: processing differences among regular allomorphs in a speeded inflection task

with Sofia Rest[☆] and Colin Wilson. At Human Sentence Processing Conference (2023).

Fellowships & Awards

SSHRC Doctoral Fellowship	2024-2026
Social Science and Humanities Research Council of Canada	\$80,000 CAD
Paul McFetridge Memorial Scholarship in Linguistics	2020
SFU VPR Undergraduate Student Research Award	2020
SFU Big Data Undergraduate Student Research Award	2019
SFU Undergraduate Open Scholarship	2017-2021

Teaching

Instructor for How to build a grammar: a crash course on linguistic fieldwork	<i>Intercession</i> 2025
◇ Designed the course from scratch, with a hands-on introduction to linguistics in mind. (6 of 7 students are non-CogSci majors)	
◇ Students worked with a native consultant of Tamil to infer its grammar.	
◇ Discussed ethical and methodological concerns with fieldwork research.	
Teaching assistant for Semantics I	<i>Fall 2023</i>
◇ Guest lecture on plurality and number.	Kyle Rawlins
Teaching assistant for Phonology I	<i>Spring 2023</i>
◇ Guest lecture on language production and the competence/performance distinction.	Colin Wilson
Teaching assistant for Bayesian Inference	<i>Fall 2022.</i> Colin Wilson
Teaching assistant for Cognitive Neuroscience	<i>Spring 2022.</i> Mick Bonner

Mentorship & Service

Mentor to Sofia Rest (Cognitive Science & Computer Science major, 2025)

- ◇ Awarded Provost’s Undergraduate Research Award for joint work on speech production.
- ◇ Developed a speeded wug inflection study and collected data together.
- ◇ Coached on experimental design and methods for analyzing speech data.

Department lead for PhD application mentorship program

- ◇ Co-managed a yearly program where we provide application advice to prospective PhD students with priority to underrepresented minority students.
- ◇ ~50 (2022), ~60 (2023), and ~80 (2024) prospective students mentored with help from graduate students throughout the department.

Department representative at the JHU Graduate Representative Organization

Technical skills

Behavioural experiments: onset latency experimental setup in-person and online, jsPsych ⇒ JATOS ⇒ MTurk/Prolific pipeline, automated analysis of speech data.

Statistical/experimental analysis: LMEMs, Bayesian statistics (Stan), EEG analysis pipeline.

Machine learning: linear encoding/decoding models, deep learning frameworks (PyTorch).

Programming/data structures: fluent in Python and R, proficient in Praat, XFST, HTML/CSS.